MG-1 FLUX-MODULATED MAGNETIC GEAR STAGE

SOLUTION DESIGN v0.1 · ALL DIMENSIONS IN MILLIMETRES · THIRD-ANGLE PROJECTION · NOT FOR PRODUCTION

MG-1

FLUX-MODULATED AXIAL MAGNETIC GEAR STAGE

4.5 : 1 counter-rotating · contactless · inherent torque limit

PARAMETER	VALUE	BASIS / NOTE
HS rotor pole pairs p_in	2	4 poles x 5 blocks
LS rotor pole pairs p_out	9	18 poles x 1 block
Modulator poles n_s	11	n_s = p_in + p_out
Gear ratio G = -p_out/p_in	-4.50	sign = counter-rotation
Magnet mean radius R_m	35.0	sets block pitch
Active OD	90.0	R_m +/- 10
Air gap, each side	0.80	shim adjustable 0.6-1.2
Modulator thickness	12.0	M8 x 25 bolts, full depth
Active axial length	23.6	5 + 0.8 + 12 + 0.8 + 5
Active volume	0.150 L	pi*45^2*23.6
Predicted pull-out (LS)	1.8-3.0 Nm	12-20 Nm/L printed build
Predicted pull-out (HS)	0.40-0.67 Nm	LS / 4.5
Max input speed	3000 rpm	-> 667 rpm out
Modulator field freq	~550 Hz	eddy-loss driver, see S/7
Total magnets	38	20 HS + 18 LS

SHEET INDEX

PURPOSE

This set specifies a BENCH INSTRUMENT, not a product.
Three ports are instrumented so the energy balance
closes with no unmeasured path:

    tau_in + tau_out + tau_react = 0

P_out <= P_in at all times. Any measurement
indicating otherwise is an instrumentation error.
Gate 4 (sheet 12) is a mandatory falsification test.

SAFETY — READ BEFORE PRINTING

PETG or ASA only. PLA creeps at 55 C; a crept
air gap is a rotor crash.
Polycarbonate burst shroud mandatory above 500 rpm,
interlocked into the enclosure chain.
Magnet pockets must be KEYED so only one polarity
orientation will seat. Polarity errors are invisible
after glue-up and are the #1 build failure.
Use the threaded-rod jig. Never hand-mate rotors.

THEORY OF OPERATION

Governing relations (Atallah-Howe topology):

```
n_s   =  p_in + p_out          coupling condition
G_r   = -p_out / p_in          modulator fixed
w_mod * n_s = w_hs * p_in + w_ls * p_out
```

Torque is capped by pull-out. Exceeding it slips
harmlessly - it cannot shear a tooth. That slip
limit is the safety primitive this build exists
to characterise.

CBJG LLC · BENCHTOP MOTOR PROGRAM				
DRAWING SET — COVER & PARAMETERS			MG-1	
Flux-modulated axial magnetic gear, benchtop	SCALE NTS	REV 0.1		SHEET 1/15

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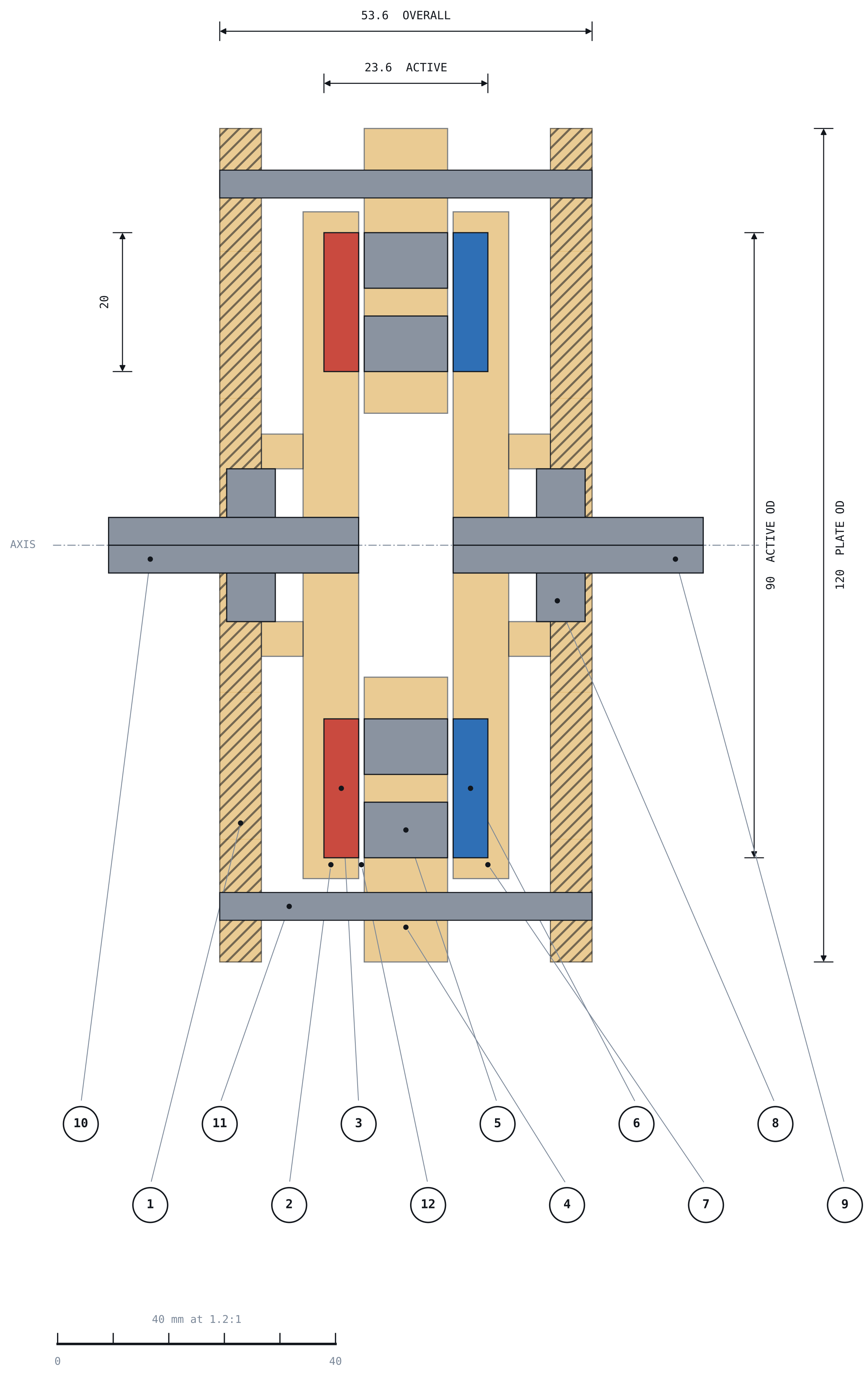

IT	DESCRIPTION
1	FRAME PLATE A PETG/ASA 6 thk
2	HS ROTOR CARRIER 8 thk (2 pole pairs)
3	HS MAGNETS 20 off N42 20x10x5
4	MODULATOR PLATE 12 thk FIXED TO FRAME
5	POLE PIECES M8x25 low-carbon steel 22 off
6	LS MAGNETS 18 off N42 20x10x5
7	LS ROTOR CARRIER 8 thk (9 pole pairs)
8	608ZZ BEARING 8x22x7 in metal sleeve
9	LS SHAFT d8 -> torque cell -> load
10	HS SHAFT d8 -> prime mover
11	M5 TIE ROD + ALUMINIUM STANDOFF 4 off
12	AIR GAP 0.80 EACH SIDE shim-set, 2 places

CRITICAL DESIGN NOTES

GAP-CRITICAL STACK MUST BE METAL:
aluminium standoffs between frame plates,
metal bearing sleeves in printed seats.
Printed material carries no gap-defining dimension.

AXIAL THRUST is large and static. Rotors are pulled toward the modulator from both sides. Size bearings for THRUST, not radial load.

Assemble only with the threaded-rod jig (sheet 3). An unrestrained rotor will free-fly into the modulator and shatter. Fingers do not survive this.

AIR GAP CONTROL

Nominal 0.80 per side. Shim stack 0.1 allows sweep at 0.60 / 0.80 / 1.00 / 1.20 for the Gate 1 torque-vs-gap curve. Torque falls steeply with gap - this sweep is the single most informative test in the programme.

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GENERAL ASSEMBLY — AXIAL SECTION				MG-1
Section A-A through shaft axis	SCALE 1.2:1	REV 0.1		SHEET 2/15

MG-1 FLUX-MODULATED MAGNETIC GEAR STAGE

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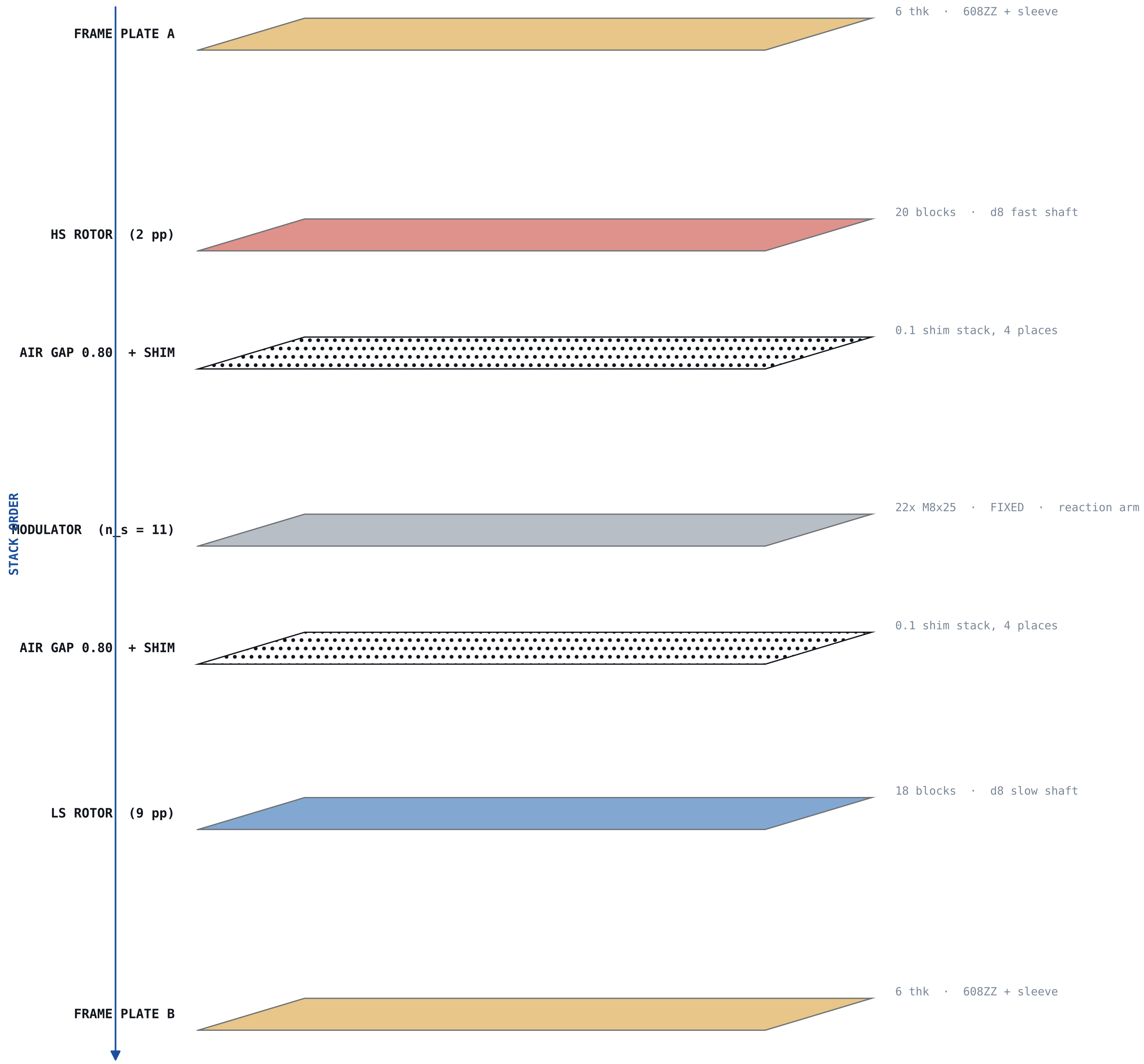

ASSEMBLY SEQUENCE

1 Print all parts in PETG/ASA. Verify pocket depth 5.0 +0.1/-0.0 with a magnet slug BEFORE bulk seating.
2 Seat magnets DRY into keyed pockets. Confirm the alternating N/S map (sheets 4-5) with a field probe or a compass on every single block.
3 Only then bond with CA. Cure fully - 24 h.
4 Press bearing sleeves, then 608ZZ into both plates.
5 Bolt 22x M8x25 into the modulator plate. Torque to snug only; the plate is plastic.
6 Fit the modulator to the 4 tie rods with standoffs and the 0.1 shim stacks. This sets both air gaps.
7 JIG: run 4 long M5 threaded rods through the tie holes. Walk each rotor in by turning nuts evenly, diagonal pairs, quarter turn at a time.
8 Replace threaded rods with final standoffs.
9 Fit the polycarbonate burst shroud and wire the interlock BEFORE first spin.

HAZARDS DURING ASSEMBLY

Never bring a rotor toward the modulator by hand.
Attraction rises steeply and closes the last few mm faster than you can react.
Keep the shroud on for every powered run.
Treat loose N42 blocks as projectiles and pinch hazards. Store them separated in a keeper.
Neodymium chips are sharp and the dust is flammable.

FIRST CHECK — RATIO BY HAND

Hand-turn the HS shaft with the LS shaft free.

4.5 input revolutions -> 1.0 output revolution
directions MUST be opposite

If the ratio is wrong, a magnet polarity is inverted.
Stop. Do not proceed. Rebuild the offending rotor.
This is Gate 0 and it is free to run.

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EXPLODED STACK & ASSEMBLY SEQUENCE			MG-1	
Stacking order, polarity keying, jig procedure	SCALE NTS	REV 0.1		SHEET 3/15

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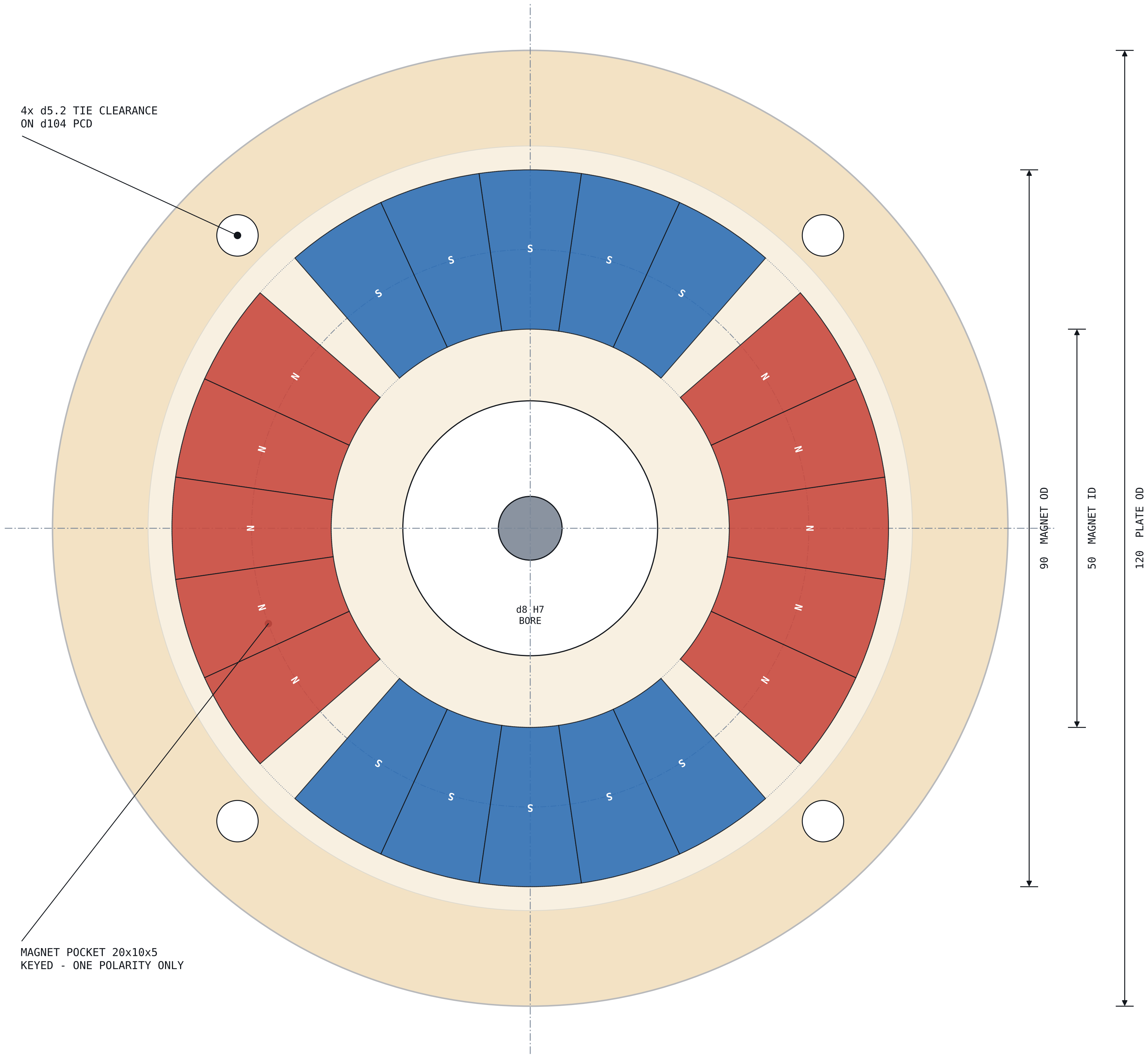

ITEM	VALUE
Pole pairs	2
Poles	4
Magnet blocks	20
Blocks per pole	5
Pole pitch	90 deg
Block angular width	16.43 deg
Pole arc / pole pitch	0.913
Mean radius R_m	35.0
Carrier thickness	8.0
Pocket depth	5.0 +0.1/-0.0
Bore	d8 H7 + keyway

WHY FIVE BLOCKS PER POLE

Each of the 4 poles is built from FIVE 10 mm blocks laid side by side, all the same polarity. Five blocks span 82.1 deg of a 90 deg pole pitch - a 0.913 pole-arc ratio, close to the optimum for this topology.

Using one wide arc magnet would be better but arc magnets are expensive and slow to source. Blocks are off-shelf and cheap.

All five blocks in a pole face the SAME way. Adjacent poles alternate. Getting one block backwards inside a pole group is the easiest mistake to make and the hardest to see. Probe every block.

PRINT SETTINGS

Material   PETG or ASA. NOT PLA.
Layer   0.16 mm, 0.4 nozzle
Walls   4 perimeters minimum
Infill   60 % gyroid, 100 % under pockets
Orientation Flat on bed, pockets UP
Supports   None required
Check pocket depth with a slug before bulk seating.

CBJG LLC · BENCHTOP MOTOR PROGRAM				
HS ROTOR PLATE — 2 POLE PAIRS			**MG-1**	
High-speed rotor, 4 poles x 5 blocks = 20 magnets	SCALE 1.5:1	REV 0.1		SHEET 4/15

MG-1 FLUX-MODULATED MAGNETIC GEAR STAGE

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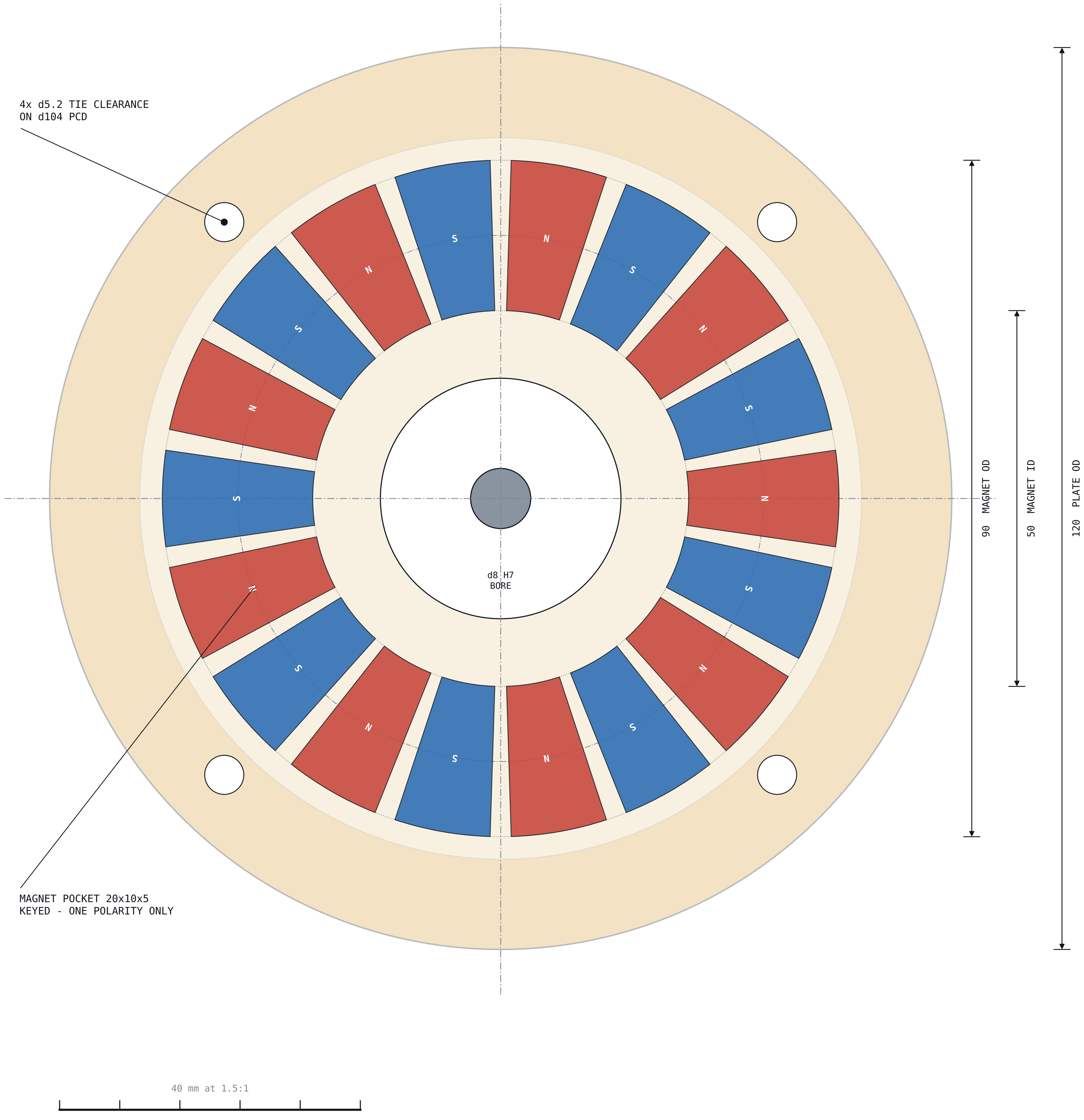

ITEM	VALUE
Pole pairs	9
Poles	18
Magnet blocks	18
Blocks per pole	1
Pole pitch	20 deg
Block angular width	16.43 deg
Pole arc / pole pitch	0.821
Mean radius R_m	35.0
Carrier thickness	8.0
Pocket depth	5.0 +0.1/-0.0
Bore	d8 H7 + keyway

LS ROTOR NOTES

18 blocks, strict N-S-N-S alternation, one block per pole. Block width 10 mm against a 12.22 mm pole pitch gives a 0.821 pole-arc ratio and a 2.2 mm inter-magnet gap at R_m - enough for print tolerance and adhesive.

This is the LOW-SPEED, HIGH-TORQUE port. It carries 4.5x the HS shaft torque, so the hub-to-shaft joint is the mechanically critical feature on this part.

Do not rely on a printed keyway alone. Use a metal hub insert or a through-pin at the shaft.

PRINT SETTINGS

Material	PETG or ASA. NOT PLA.
Layer	0.16 mm, 0.4 nozzle
Walls	4 perimeters minimum
Infill	60 % gyroid, 100 % under pockets
Orientation	Flat on bed, pockets UP
Supports	None required

Check pocket depth with a slug before bulk seating.

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LS ROTOR PLATE — 9 POLE PAIRS			MG-1	
Low-speed rotor, 18 poles x 1 block = 18 magnets	SCALE 1.5:1	REV 0.1		SHEET 5/15

MG-1 FLUX-MODULATED MAGNETIC GEAR STAGE

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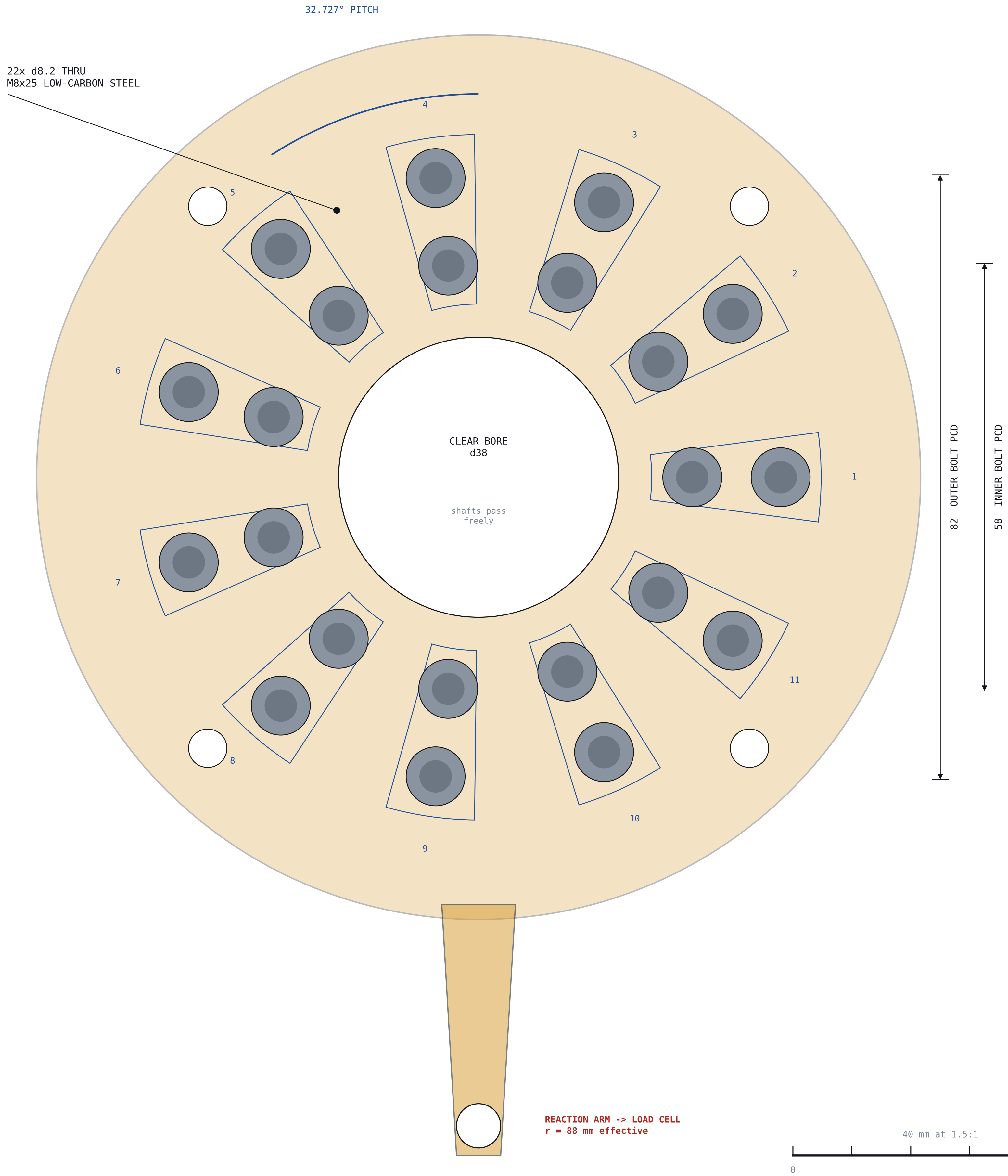

ITEM	VALUE
Modulator poles n_s	11
Angular pitch	32.727 deg
Bolts per pole	2 (radial pair)
Total bolts	22
Inner bolt PCD	d58
Outer bolt PCD	d82
Plate thickness	12.0
Bolt spec	M8 x 25, grade 4.8
Bore	d38 clear
Mounting	4x M5 tie rod, FIXED
Field frequency @3000rpm	~550 Hz

WHY 11

n_s = p_in + p_out = 2 + 9 = 11. This is the whole trick. The steel is NOT structural - it spatially modulates the field so a 2-pole-pair rotor couples to a 9-pole-pair rotor. Get the count wrong and torque collapses to near zero.

TWO bolts per pole, at d58 and d82, give radial coverage across the 20 mm magnet band. A single M8 covers only 8 mm of it and wastes most of the magnet. Both bolts sit at the SAME angular position - they are one pole piece, not two.

EDDY LOSS & THE V1/V2/V3 PATH

Solid M8 bolts are electromagnetically poor: eddy loss scales as d^2 * f^2, and at 3000 rpm these poles see roughly 550 Hz.

```
V1  22x solid M8          baseline, instrument it
V2  11x 7-off M3 bundles  d/2.7 -> loss ~7x lower
V3  stacked 0.35 laminations   approaches literature
```

V1 is not a compromise - the V1/V2/V3 loss curve is the actual deliverable of this build.

CBJG LLC · BENCHTOP MOTOR PROGRAM				
MODULATOR RING — 11 POLE PIECES			**MG-1**	
Stationary flux modulator, 22 x M8 steel bolts	SCALE 1.5:1	REV 0.1		SHEET 6/15

MG-1 FLUX-MODULATED MAGNETIC GEAR STAGE

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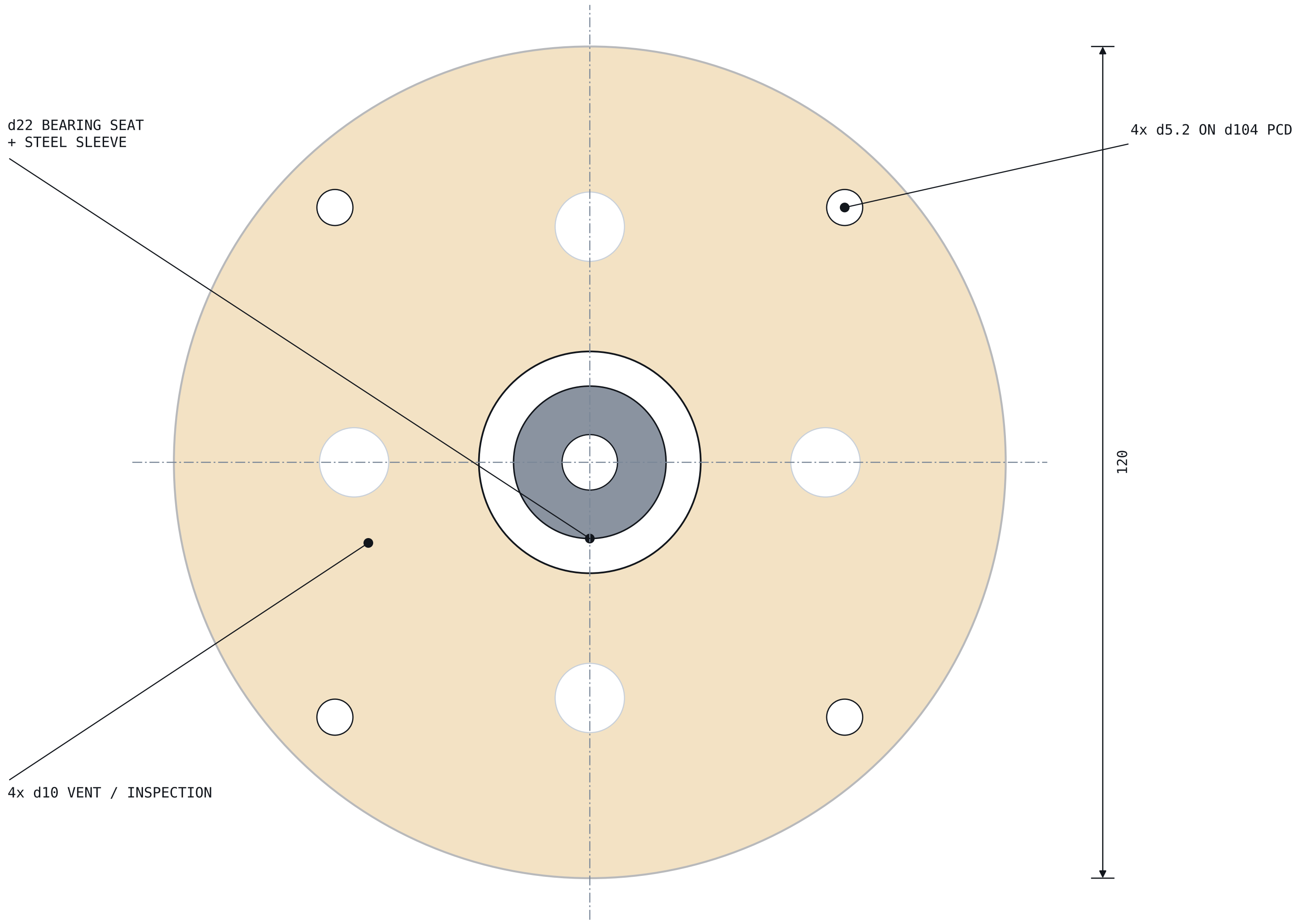
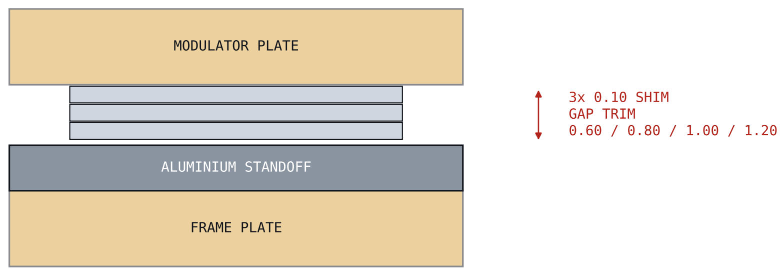
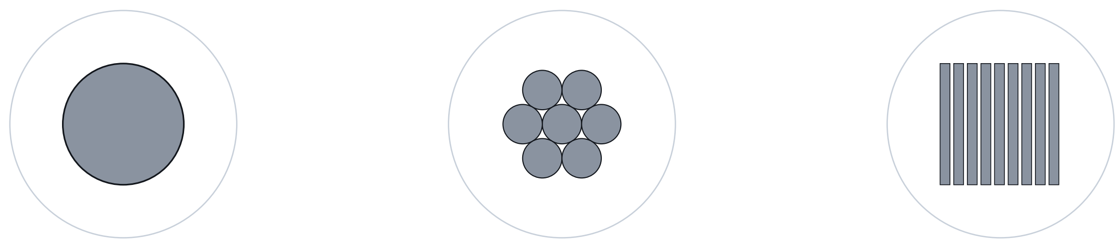
DETAIL B — FRAME PLATE (2 off, 1:1)



DETAIL C — AIR GAP SHIM STACK (6:1)



DETAIL D — POLE PIECE VARIANTS (2:1)



V1 1x M8 solid

V2 7x M3 bundle

V3 0.35 laminations

VAR	EFF. d	EDDY LOSS	BUILD
V1	8.0 mm	baseline (1.0x)	off-shelf bolts, 1 h
V2	3.0 mm	~0.14x	M3 rods + drill jig
V3	0.35 mm	~0.002x (ideal)	EDM / stacked laser-cut

FEATURES & FITS

Bearing seats: d22 H7 in a STEEL SLEEVE, sleeve bonded into the printed boss. Never run a 608 directly in plastic - the seat creeps and the gap follows it.

Tie rods: 4x M5 stainless, full length 53.6, with aluminium standoffs cut to length on a lathe or bought precision-ground.

Vent holes: 4x d10 for convective cooling of the modulator. Sized so no finger fits.

Thermistor: 10k NTC bonded to an outer M8 bolt head, wires routed through a vent.

Reaction arm: integral to the modulator plate, 88 mm effective radius to the cell. tau_react = F_cell x 0.088 m

MEASUREMENT INTEGRITY

Do not print the reaction arm hollow. It is a measurement datum - deflection is measurement error. Solid infill, 6 perimeters, or use an aluminium bar.

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FRAME PLATE, SHIM DETAIL & MODULATOR VARIANTS MG-1

Detail views B, C, D | SCALE AS NOTED | REV 0.1 | SHEET 7/15

MG-1 FLUX-MODULATED MAGNETIC GEAR STAGE

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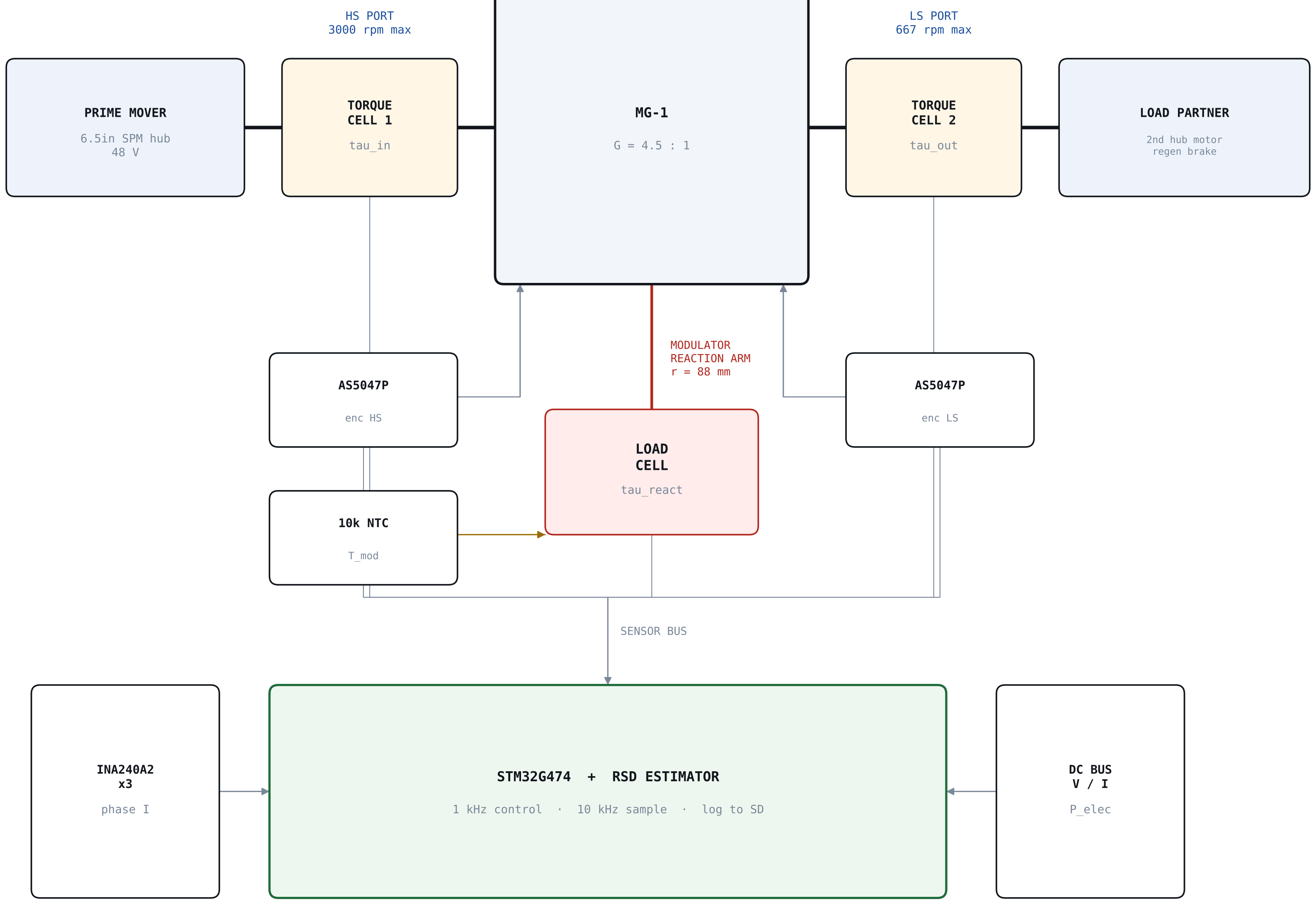

CHANNEL LIST

Ch	Quantity	Sensor	Purpose
1	tau_in	inline rotary	mechanical power in
2	tau_out	inline rotary	mechanical power out
3	tau_react	arm + load cell	CLOSES the balance
4	w_hs	AS5047P	slip numerator
5	w_ls	AS5047P	slip denominator
6	T_mod	10k NTC on bolt	eddy heating
7	V_bus,I_bus	INA240A2	electrical input

WHY THE THIRD PORT MATTERS

Three ports measured means no unmeasured path:

$$\tau_{in} + \tau_{out} + \tau_{react} = 0 \quad \text{(static)}$$

$$P_{in} - P_{out} = P_{loss} \geq 0 \quad \text{(always)}$$

Most rigs measure two ports and infer the third. That inference is exactly where a spurious over-unity reading hides. Measuring all three removes the hiding place - which is the point.

RSD ROLE — ESTIMATOR ONLY

SLIP OBSERVER
- compare w_ls against w_hs / 4.5
- persistent deviation => pull-out slip => latch
- a contactless gearbox that reports its own overload
- with NO torque sensor in the loop

THERMAL VIRTUAL SENSOR
- estimate T_mod from (speed, transmitted torque)
- cross-check against the NTC
- model-vs-sensor divergence is itself a fault signal

RSD IS NOT IN THE SAFETY PATH. Slip detection is an observation. Only the hardware chain (sheet 10) has authority to remove torque.

CALIBRATION DISCIPLINE & CH 3 RANGE

Calibrate both torque cells against the SAME dead-weight reference before Gate 3, then SWAP them between ports and repeat. Any asymmetry that appears on swap is calibration error, not physics. This is the core of Gate 4.

Ch 3 range: the ring carries (n_s / p_in) x tau_HS = 5.5 x 0.53 = 2.93 N.m, i.e. 33.3 N at the 88 mm arm - the LARGEST torque in the machine, larger than either shaft. Specify the cell at 50 N full scale.

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THREE-PORT INSTRUMENTATION BLOCK DIAGRAM | MG-1

Closed energy balance across three ports | SCALE NTS | REV 0.1 | SHEET 8/15

MG-1 FLUX-MODULATED MAGNETIC GEAR STAGE

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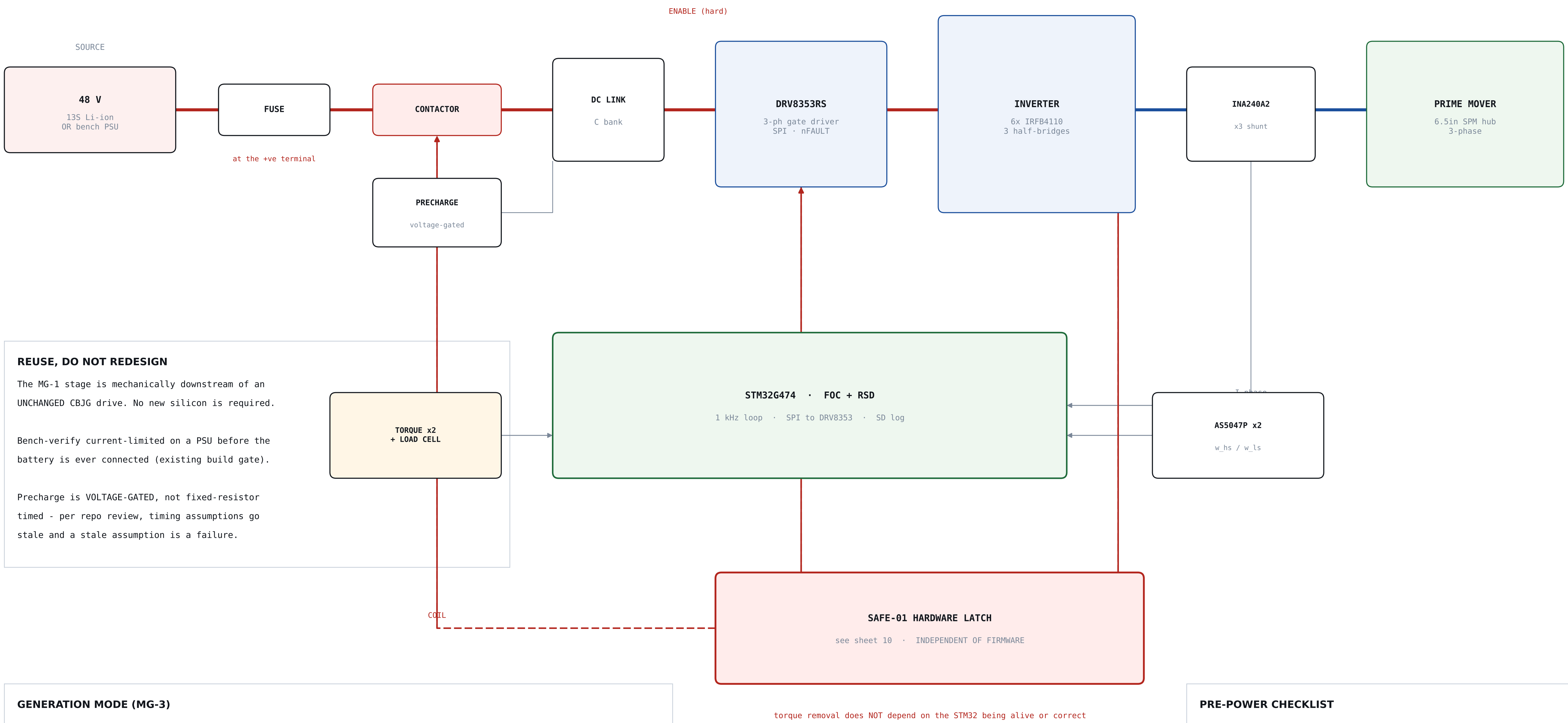

REUSE, DO NOT REDESIGN

The MG-1 stage is mechanically downstream of an UNCHANGED CBJG drive. No new silicon is required.

Bench-verify current-limited on a PSU before the battery is ever connected (existing build gate).

Precharge is VOLTAGE-GATED, not fixed-resistor timed - per repo review, timing assumptions go stale and a stale assumption is a failure.

GENERATION MODE (MG-3)

For MG-3 generation mode the same bridge becomes an ACTIVE RECTIFIER: the load partner drives the LS port, MG-1 steps speed UP into the machine, and the inverter regenerates into the 48 V bus.

A regenerating bridge will push bus voltage ABOVE 48 V. Provide a brake chopper or a sized dump resistor before any generation test. The capacitor bank is not a sink.

PRE-POWER CHECKLIST

Fuse at the +ve terminal
Contactor before the DC link
Voltage-gated precharge
Enclosure interlock in the latch
E-stop in the latch
Burst shroud fitted
Dump resistor for regen
Current-limited first run

CBJG LLC · BENCHTOP MOTOR PROGRAM				
POWER & MOTOR CONTROL SCHEMATIC			**MG-1**	
48 V bus, 3-phase inverter, existing stack	SCALE NTS	REV 0.1		SHEET 9/15

MG-1 FLUX-MODULATED MAGNETIC GEAR STAGE

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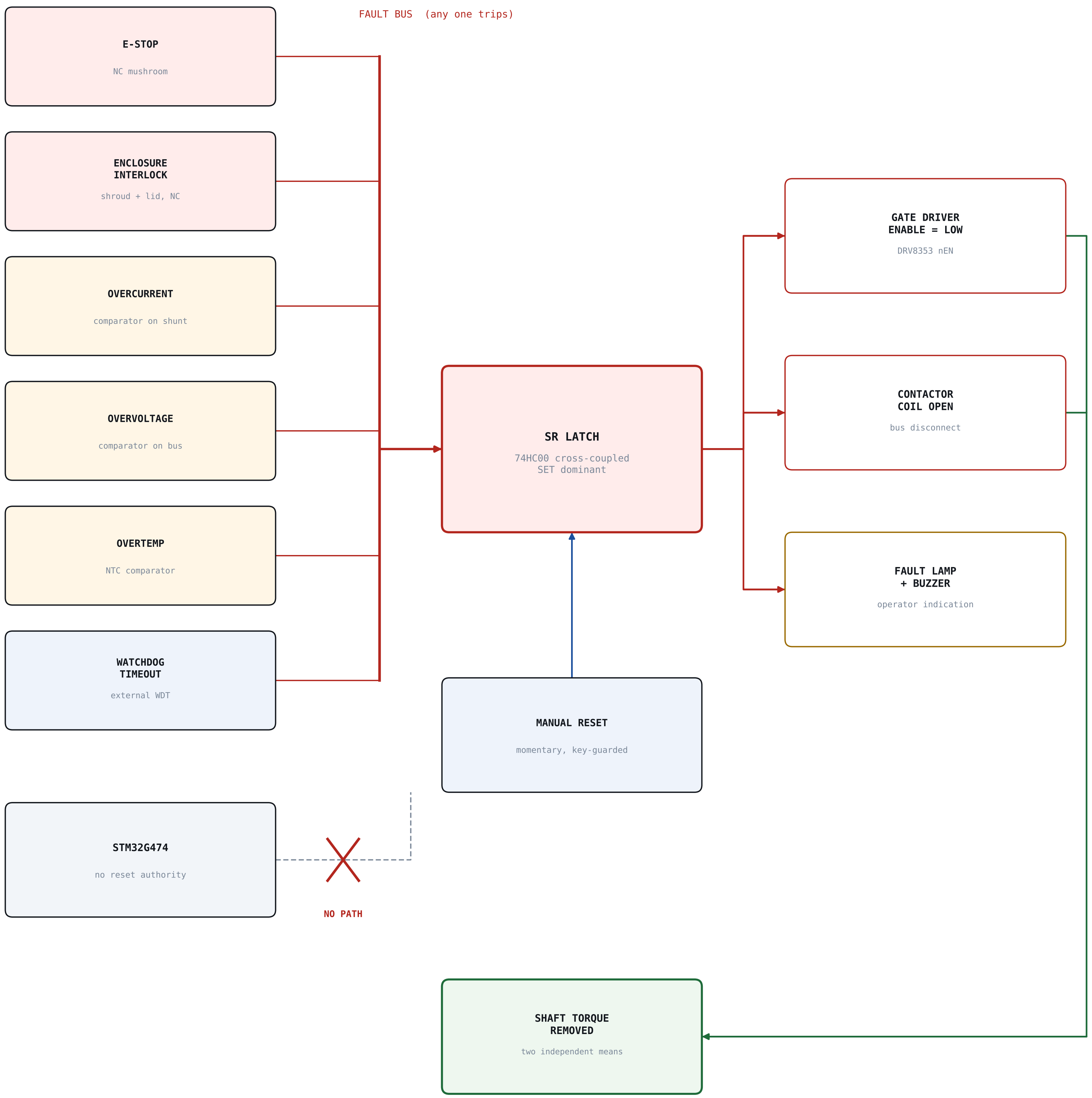

OPERATING PRINCIPLE

1 Any fault source latches the chain in HARDWARE.
2 The latch removes gate-driver enable AND opens the contactor. Two independent means of torque removal.
3 The latch holds with the STM32 dead, hung, or running wrong firmware.
4 Only a deliberate MANUAL reset clears it. Firmware has no reset path - not a GPIO, not a register.
5 Reset is refused while any fault source is still asserted (SET-dominant latch).

CONTROL POINTS & OPEN GATES

Issue #14 is CANONICAL for reset semantics and SAFE-01 behaviour. Firmware issue #13 DEFERS to it.
Enclosure interlock coverage across issues #5/#6/#7/#13/#14 must be resolved BEFORE hardware spin-up.

MG-1 additions to the existing chain:

- burst shroud position switch -> enclosure interlock
- modulator NTC -> overtemp comparator
- RSD slip flag -> LOG AND ALARM ONLY, never a trip

THE TORQUE FUSE — WHY MG-1 EARNS ITS SLOT

MG-1 adds a SECOND, purely physical torque limit that sits downstream of everything on this sheet:

pull-out slip = mechanical torque ceiling

No firmware fault, no gate-driver failure and no shorted MOSFET can push more than pull-out torque through the shaft. SAFE-01 acts at the inverter; the magnetic gear acts at the shaft.

Record the measured Gate 1 pull-out figure as a SPECIFIED SAFETY PARAMETER, not a test result.

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SAFE-01 — SAFE TORQUE OFF CHAIN			MG-1	
Hardware latched, firmware independent	SCALE NTS	REV 0.1		SHEET 10/15

MG-1 FLUX-MODULATED MAGNETIC GEAR STAGE

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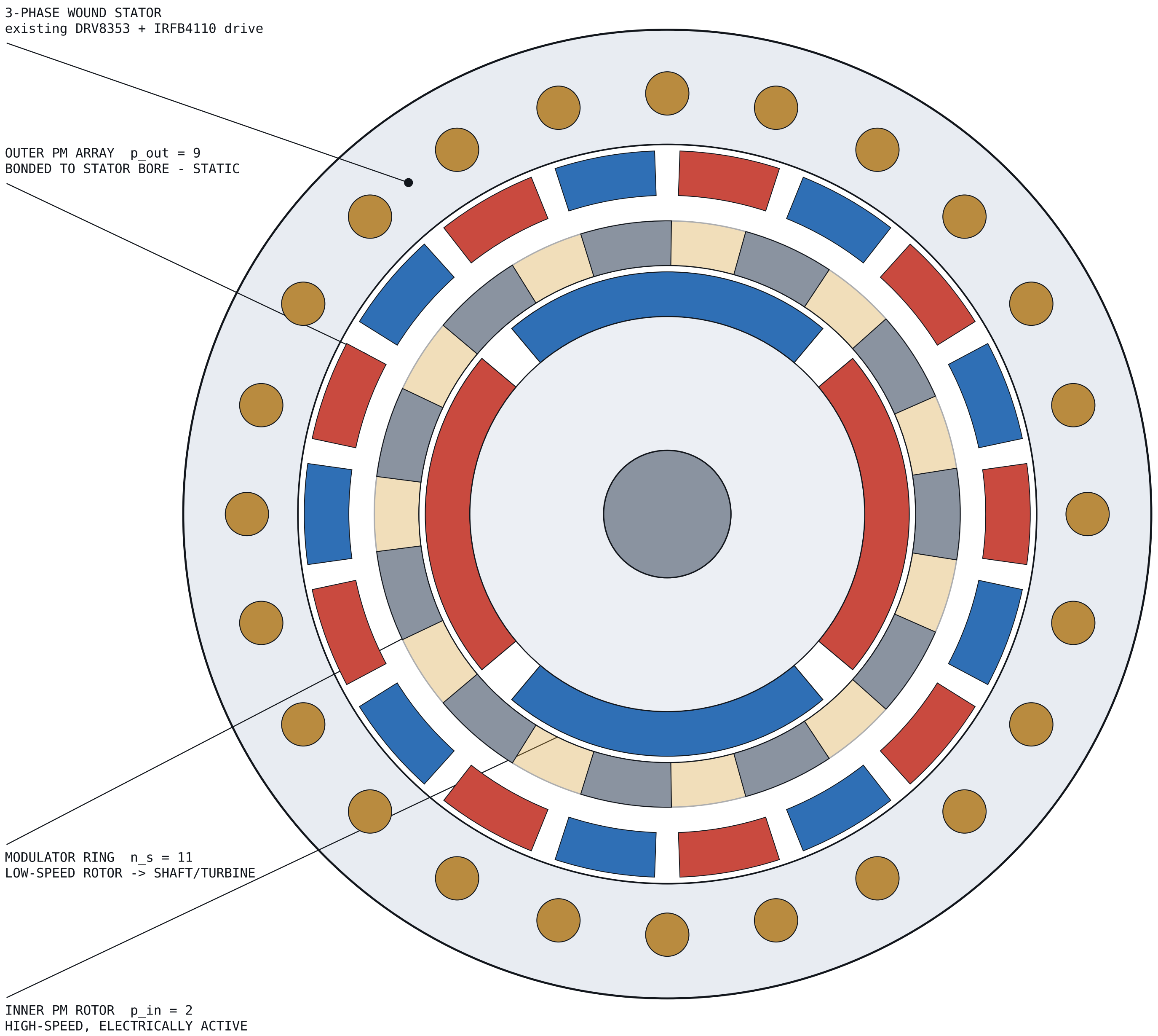

WHAT CHANGES FROM MG-1

MG-1 is a gear. MG-2 merges the gear WITH the machine: wrap a wound stator around the outer magnet array and the gear's high-speed rotor becomes the motor rotor. One housing, three concentric parts, no gear teeth.

Reference performance for this architecture:

- 60-90 Nm/L natural air cooling (vs 10-30 conv. PM)
- 110 kNm/m^3 for large machines
- torque ripple < 0.3 % , power factor > 0.9
- <60 % the size and mass of a PM machine

CONTROL IMPLICATIONS

The machine commutates against p_in = 2.
The shaft turns at p_out = 9.

Consequences for the existing firmware:

- FOC electrical angle tracks the INNER rotor
- encoder must be on the inner rotor, or the inner angle must be reconstructed from the modulator angle plus the ratio
- RSD's estimator carries G = 4.5 EXPLICITLY; a hidden ratio is a hidden fault mode
- slip between ports is now an ELECTRICAL fault signature, not just a mechanical one

PATH TO GENERATION (MG-3)

MG-3: run MG-2 backwards. Drive the slow port with the hoverboard motor, rectify the stator output. Because all three ports are instrumented, GEAR loss and ELECTRICAL loss separate cleanly instead of collapsing into one lumped efficiency number.

That separation is the reason to build the rig at all.

Grid & Chain crossover: pull-out slip is a RIDER SAFETY property - a seized wheel or a controller fault physically cannot exceed pull-out torque.

STATUS — CONCEPT ONLY

Proportions are indicative only. This sheet is a concept, not a manufacturing drawing - slot count, winding layout, back-iron thickness and gap tolerances all require FEA before any part is cut.

Do not build MG-2 until MG-1 Gates 0-2 have passed. The gear must be characterised before it is buried inside a machine where its behaviour cannot be seen.

CBJG LLC · BENCHTOP MOTOR PROGRAM				
MG-2 PSEUDO DIRECT DRIVE — CONCEPT SECTION			MG-2	
Radial-flux PDD: stator wraps the gear	SCALE CONCEPT	REV 0.1		SHEET 11/15

MG-1 FLUX-MODULATED MAGNETIC GEAR STAGE

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GATE 0

STATIC, NO POWER

- [ ] Hand-turn input, count revolutions
- [ ] 4.5 in : 1 out, OPPOSITE directions
- [ ] Wrong ratio = inverted magnet. STOP.

GATE 1

STATIC PULL-OUT

- [ ] Lock output, torque wrench on input
- [ ] Record breakaway torque + angle, 10x
- [ ] Repeatable within +/-10 %
- [ ] Sweep gap 0.60/0.80/1.00/1.20

GATE 2

CURRENT-LIMITED SPIN

- [ ] Bench PSU only. NO battery.
- [ ] 100 -> 500 rpm, 20 min soak each
- [ ] Stop if printed frame exceeds 50 C

GATE 3

LOADED EFFICIENCY MAP

- [ ] Shroud fitted, interlock live
- [ ] Sweep speed x torque
- [ ] eta = P_out / P_in across the map

GATE 4

FALSIFICATION — MANDATORY

- [ ] Three-port balance, cells cross-calibrated
- [ ] SWAP cells between ports, repeat
- [ ] Required result: NO excess energy
- [ ] Document the uncertainty band

EXIT CRITERIA — ALL MUST HOLD

CRITERION	TARGET
Ratio verified by hand	4.50 : 1, opposite
Pull-out repeatability	+/- 10 %
Efficiency, V1, 40-80 % pull-out	>= 80 %
Thermal runaway at rated speed	none
Frame temperature	< 50 C
Energy balance closure	within uncertainty
Shroud + interlock	fitted & proven

RISK	IMPACT	MITIGATION
Magnet polarity error	near-zero torque	keyed pockets, Gate 0
Frame creep -> gap collapse	rotor crash	PETG/ASA, metal stack
Eddy heating in solid bolts	eta collapse	NTC, V2/V3 planned
Uncontained rotor burst	injury	PC shroud + interlock
Cell miscalibration	false over-unity	3-port, swap at Gate 4
Pinch during assembly	injury	threaded-rod jig
Regen overvoltage (MG-3)	bus damage	dump resistor
'Free energy' scope creep	credibility	Gate 4 contractual

PROGRAMME HOOKS

Open a GitHub issue in giomj/dev for MG-1, cross-referencing the SAFE-01 interlock chain (#14 canonical). Record the measured Gate 1 pull-out torque as a SPECIFIED SAFETY PARAMETER once it exists. Resolve enclosure interlock coverage across #5/#6/#7/#13/#14 before spin-up. Execute the Phase 0 motor purchase before spending further on silicon.

P_out <= P_in. Always. Gate 4 exists to prove it on this rig, with these sensors, and to publish the negative result. That is what makes the rest credible.

PHASE PLAN

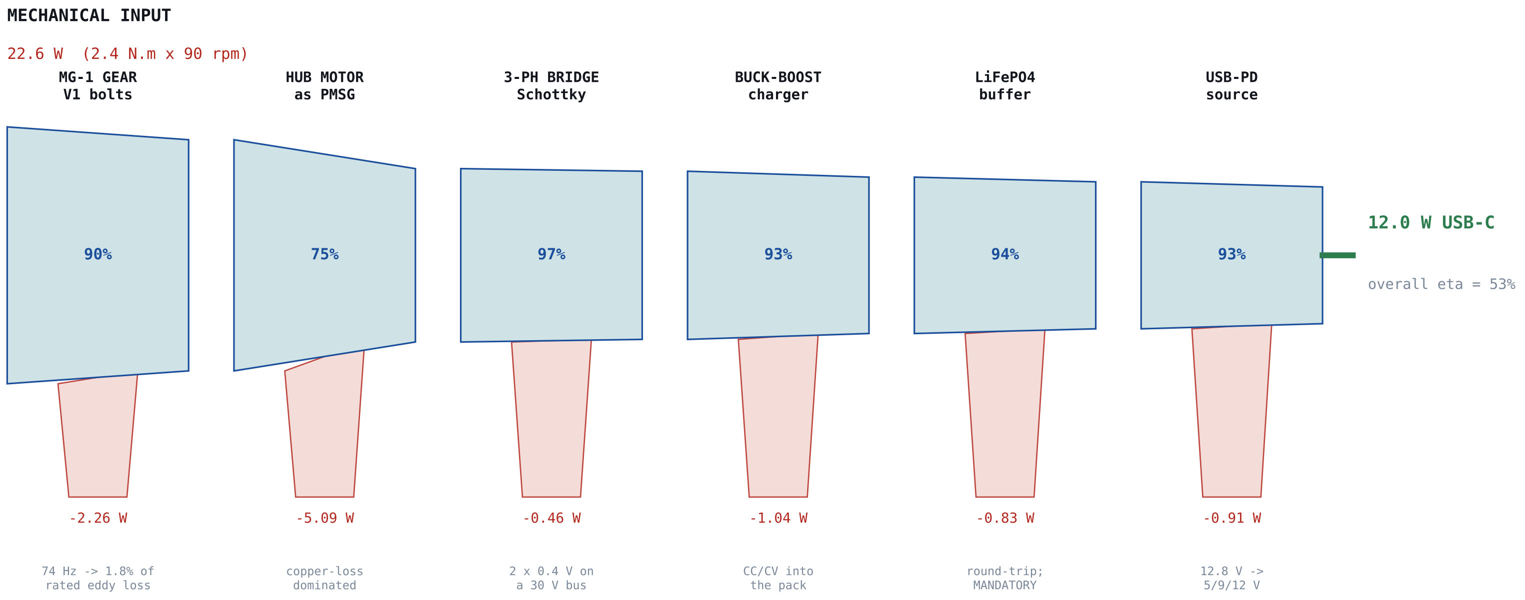
P0 Model analytical/FEA sizing, pull-out vs gap
P1 Build V1 print, key, seat, bond, jig-assemble
P2 Instrument 3-port sensing, slip observer, NTC
P3 Characterise full map + Gate 4 falsification
P4 V2 / V3 segmented rods, then laminations
P5 Integrate standard CBJG dyno reduction module

Then MG-2 (stator) -> MG-3 (generation) -> demonstrator.

CBJG LLC · BENCHTOP MOTOR PROGRAM				
BUILD, GATE & VERIFICATION CHECKLIST			MG-1	
Gates 0-4, exit criteria, risk register	SCALE NTS	REV 0.1		SHEET 12/15

MG-1 FLUX-MODULATED MAGNETIC GEAR STAGE

SOLUTION DESIGN v0.1 · ALL DIMENSIONS IN MILLIMETRES · THIRD-ANGLE PROJECTION · NOT FOR PRODUCTION



WHAT THIS SHEET DOES NOT CLAIM

P_out <= P_in AT ALL TIMES. Every stage on this sheet removes energy. Nothing on this sheet adds energy. The gear is a TRANSFORMER of speed and torque, not a source.

GATE 4 falsification test (sheet 12) is contractual: measure all three ports, cross-calibrate the cells, then SWAP them between ports. Required result is NO excess energy.

WHY 12 W AND NOT 54 W

The operator is NOT the limit. Sustained comfortable one-hand cranking measures 54 +/- 14 W (Jansen & Slob, ICED 2003). MG-1's low-speed pull-out torque caps mechanical input at ~23 W at 90 rpm.

The torque fuse that makes this machine safe is the same mechanism that caps its harvest power. That is a design trade, stated openly, not a defect.

More power therefore means MORE ACTIVE VOLUME, not a better winding: tau scales with volume at 12 - 20 N.m/L for this topology.

EDDY-LOSS REGIME

Crank speeds put the modulator field at only 74 Hz against ~550 Hz at rated speed. Eddy loss goes as f^2, so it is ~1.8% of the rated figure: the V1 solid M8 pole pieces are entirely adequate HERE.

The V2 / V3 laminated sweep matters at 3000 rpm, not in the charger duty cycle.

OPERATING POINT	tau_LS	P_mech	P_USB	eta
Pull-out low, 12 N.m/L	1.8 N.m	17.0 W	9.0 W	53%
DESIGN POINT	2.4 N.m	22.6 W	12.0 W	53%
Pull-out high, 20 N.m/L	3.0 N.m	28.3 W	15.1 W	53%

LOAD	CAPACITY	TIME AT 12 W
Phone, 4000 mAh	15.4 Wh	1.3 h
Tablet	25 Wh	2.1 h
20 000 mAh bank	74 Wh	6.1 h

ACTIVE VOL	tau_LS BAND	P_mech at 90 rpm
0.15 L (MG-1)	1.8 - 3.0 N.m	17 - 28 W
0.30 L	3.6 - 6.0 N.m	34 - 57 W
0.50 L	6.0 - 10 N.m	57 - 94 W
1.00 L	12 - 20 N.m	113 - 188 W

CBJG LLC · BENCHTOP MOTOR PROGRAM				
MG-C1 ENERGY TRANSFORMER - POWER FLOW			MG-1	
Sankey-style budget at the 90 rpm crank design po...	SCALE NTS	REV 0.1		SHEET 13/15

MG-1 FLUX-MODULATED MAGNETIC GEAR STAGE

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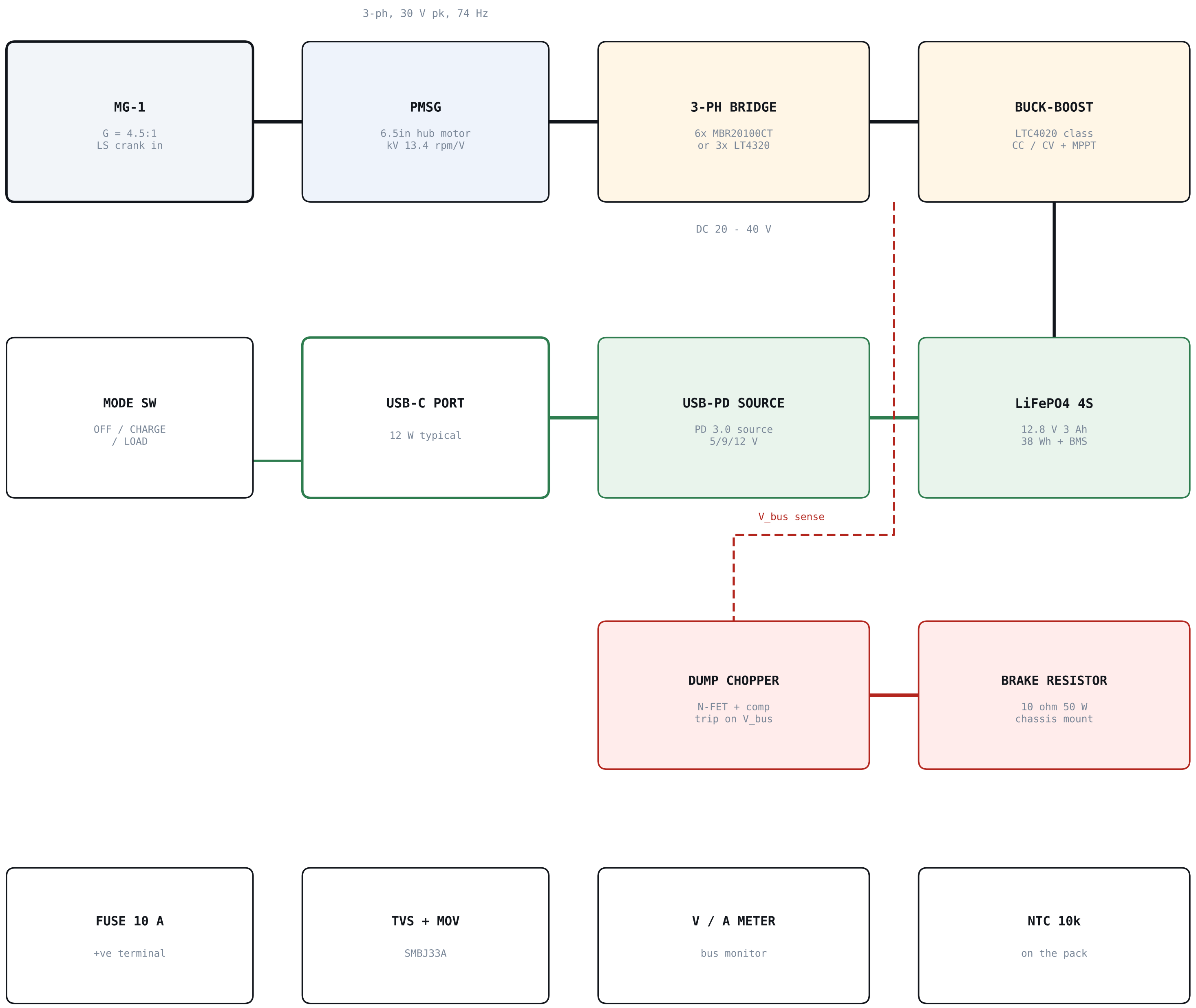

PROTECTION RAIL - fitted on the input side of the bridge, not as an afterthought

THE #1 WIRING MISTAKE

A PD 'trigger' or 'decoy' board is a SINK. It asks an upstream charger for 9 V or 12 V. It cannot supply a device. This chain needs a genuine PD SOURCE controller (MPQ5031 / TPS25751 / STUSB4761 class) or an off-the-shelf power-bank board. Fitting a trigger board here yields exactly 0 W.

WHY A BUFFER CELL

The buffer cell is MANDATORY, not an upgrade. A phone negotiates a contract and then expects a stable rail. Hand-crank output is a fluctuating source: speed varies every half revolution, and a hand pause is a total dropout.

The pack decouples them. You crank into the pack; the pack, never the crank, talks to the phone.

RECTIFIER: TIER A vs TIER B

Tier A: 6x MBR20100CT Schottky. At a 30 V bus the two 0.4 V drops cost only ~3%.

Tier B: 3x LT4320 ideal-diode bridge controllers + 6 N-FETs. AD's own 3-phase demo shows 84% for a diode bridge vs 97% ideal at 5 VAC L-N. That gain only matters if the bus falls below ~10 V, i.e. a much slower crank or a higher-kV generator.

SAFE-01 ALIGNMENT

The capacitor bank is NOT an energy sink. On rotor slip or a sudden stall the bus rises; the dump chopper and brake resistor are what absorb it.

RSD stays an estimator, fault detector and thermal virtual sensor. It is NEVER in the safety path.

CBJG LLC · BENCHTOP MOTOR PROGRAM				
MG-C1 CHARGING CHAIN - SCHEMATIC			MG-1	
PMSG to 3-ph rectifier to buck-boost to buffer to...	SCALE NTS	REV 0.1		SHEET 14/15

MG-1 FLUX-MODULATED MAGNETIC GEAR STAGE

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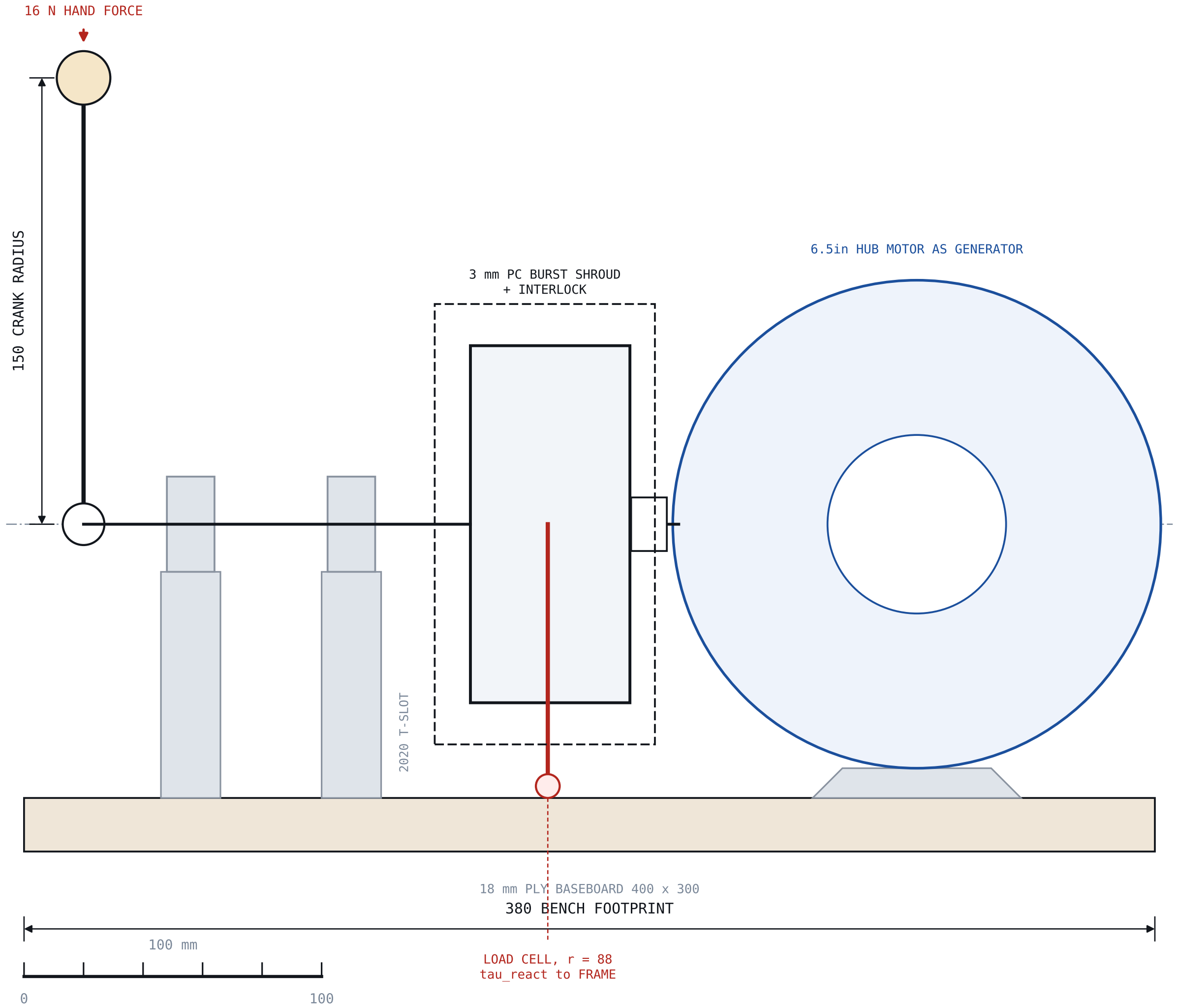

BUILD TIER	HARDWARE	DELIVERED
1 CHARGER only	$847	$1 086
2 + INSTRUMENTATION	$987	$1 265
3 + VARIANTS & SENSORS	$1 584	$2 032

BOM SECTION	SUBTOTAL	TIER
A Printed parts	$22	CHG
B Magnets & bonding	$99	CHG
C Metal, fasteners, shafts	$125	CHG/OPT
D Drive, crank, generator	$215	CHG
E Power conversion chain	$193	CHG/OPT
F Instrumentation	$660	INST/OPT
G Consumables & tooling	$271	CHG/OPT

COST NOTES

Delivered totals add 8% freight, 8.25% tax and 12% contingency. All rates are live cells in the workbook - change them, do not trust them.

Tier 3 is dominated by ONE line: the pair of rotary torque sensors at ~$260 each. Start on load cells and upgrade at Gate 3 only if the uncertainty band is too wide to close the balance.

C1: the 22 pole-piece bolts MUST be plain low-carbon steel. A2 / A4 stainless is austenitic and nearly non-magnetic - fit it and the gear makes no torque.

TORQUE PATH	SPEED	TORQUE	POWER
Crank / LS port IN	90 rpm	2.4 N.m	22.6 W
HS port OUT	405 rpm	0.53 N.m	20.4 W
Modulator reaction	0 rpm	2.93 N.m	0 W
USB-C port	-	-	12.0 W

tau_in + tau_out + tau_react = 0, all EXTERNAL torques about +z. The ring carries (n_s / p_in) x tau_HS = 5.5 x 0.53 = 2.93 N.m -- the LARGEST torque here. Size arm and ring bolts for it, not the HS shaft. Cell force = 2.93 / 0.088 = 33.3 N. Specify 50 N full scale. HS shaft to generator: jaw coupling, Buna insert, inside the shroud. Crank / LS side runs on two 2020-mounted pillow blocks.

CBJG LLC · BENCHTOP MOTOR PROGRAM				
MG-C1 CRANK / MOUNT GENERAL ARRANGEMENT			MG-1	
Bench layout, reaction path and BOM roll-up	SCALE 1:2.5	REV 0.1		SHEET 15/15